Figure 12.31 outlines how PCR technique works. To amplify (make large quantities) a particular DNA sequence by PCR a reaction mixture (often 100ml or less in volume) containing the following are required.

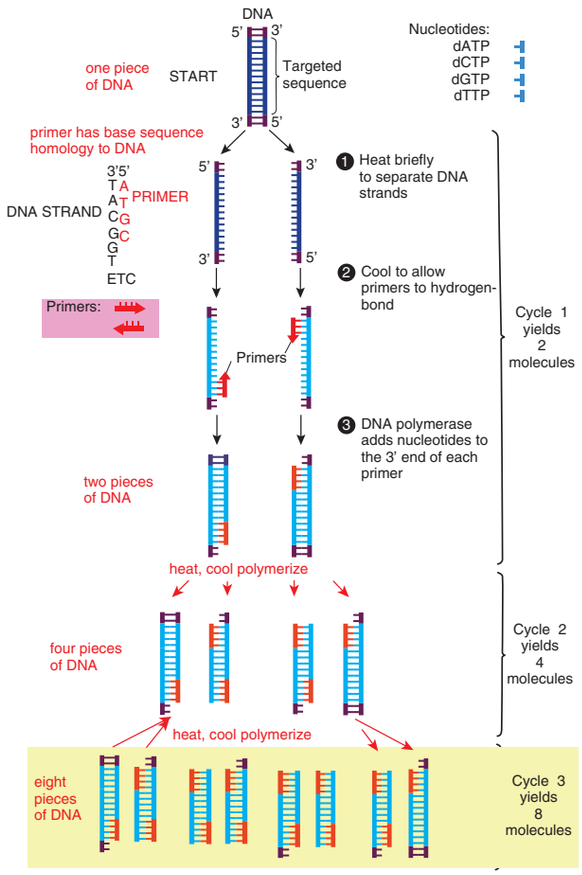


Figure 12.31: Steps in PCR

1. **Target DNA**
2. **Two primers**–These are synthetic oligonucleotides, usually about 20 nucleotides long. These are fragments with sequences identical to those flanking the targeted sequence.
3. **Thermostable DNA polymerase**– Two popular enzymes employed in the PCR technique are Taq polymerase from the thermophilic bacterium *Thermus aquaticus* and the vent polymerase from *Theromococcus litoralis*. These polymerases employed in PCR technique are able to function at high temperatures.
4. **Four deoxyribonucleoside triphosphates** (dNTPs)- dCTP, dATP, dCTTP, dTTP



Taq polymerase lacks proof reading exonuclease (3′–5′) activity which might contribute to errors in the products of PCR. Some other thermostable DNA polymerases with proof reading activity have been identified. Example: Tma DNA polymerase from *Thermotoga maritana*.

Steps in PCR

1. **Denaturation:** The target DNA containing the sequence to be amplified is heat denatured to separate its complementary strands at temperature 94 °C–95 °C.
2. **Annealing:** The temperature is lowered to 37 °C–55 °C so that the primers can hydrogen bond or anneal to the DNA on both sides of the target sequence. Because the primes are present in excess the targeted DNA strands normally anneal to the primers rather than to each other.
3. **Extension:** Heat resistant DNA polymerase extends the primers and synthesizes copies of the target DNA sequence using the deoxyribonucleoside triphosphate's at 70 °C–75 °C.

The three – step cycle (Figure 12.32) is repeated to obtain copies of target DNA in large numbers. At the end of one

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